86% practical 26/30 practical

oral 80% 18/30 oral

OBSERVATION CHECKLIST

ABERDARE TECHNICAL COLLEGE

INFORMATION COMMUNICATION TECHNOLOGY

MODULE 2, TERM 2

Unit	Computer Repair and Maintenance
Unit code	0714 351 04A
Element 3	Test Computer Component Functionality
Assessment	Practical test: Testing RAM using MemTest86

Candidate and assessor details

Candidate name	Collins Ngotho
Registration no.	1083
Station / computer no.	001
Date	2/11/2026
Assessor name	James
Start time / End time	Start 8:00am End 2PM

SECTION A: CUTTING LIST (REQUIREMENTS)

Assessor to confirm that all items are available before the assessment begins.

No.	Item	Quantity	Provided by
1	Computer to be tested (desktop or laptop) with at least one RAM module installed	1	Institution

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No.	Item	Quantity	Provided by
2	Second working computer with internet access (for downloading and preparing the USB)	1	Institution
3	Internet connection (for downloading the MemTest86 USB image)	1	Institution
4	USB flash drive of at least 1 GB (will be erased)	1	Candidate
5	MemTest86 USB image file (memtest86-usb.zip containing imageUSB.exe and memtest86-usb.img)	1	Candidate (downloaded during the test)
6	Rufus (optional alternative tool for writing the image)	1	Candidate (downloaded if needed)
7	Answer sheet	1	Institution
8	Pen	1	Candidate

SECTION B: PRACTICAL CHECKLIST (30 MARKS)

Award the full mark only when the candidate performs the step correctly without assistance. Award 0 if the step is not done or is done incorrectly.

Task 1: Preparation and download (5 marks)

No.	Observable performance criteria	Max marks	Marks awarded	Remarks
1.1	Opens a web browser and goes to the official MemTest86 site or direct download link	1	0	Opened web browser visited official memtest 86 site

TASK 2

1

No.	Observable performance criteria	Max marks	Marks awarded	Remarks
1.2	Downloads the memtest86-usb.zip file successfully	1	01	downloaded the zip file
1.3	Extracts the ZIP file using Extract All	1	01	Extracted file using Extract all
1.4	Identifies imageUSB.exe and memtest86-usb.img in the extracted folder	1	01	identified image usb exe
1.5	Records the RAM information of the test computer (total size, speed, slots used) from Task Manager > Performance > Memory	1	01	Size Recorded Speed Recorded as 2400 MTs slots used recorded as 1 of 2
	Sub-total	**5**	05	

Task 2: Creating the bootable MemTest86 flash drive (6 marks)

No.	Observable performance criteria	Max marks	Marks awarded	Remarks
2.1	Backs up or checks the flash drive and inserts it into the working computer	1	00	No Backup Performed
2.2	Runs imageUSB.exe as Administrator	1	01	Runs usb exe as administrator
3	Selects only the correct flash drive in Step 1 (verifies drive letter and size)	1	01	selected only the correct flash drive
	Selects "Write image to UFD" in Step 2	1	01	selected
	Selects memtest86-usb.img in Step 3	1	01	in step 3 selected usb image

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If the computer will not boot from the USB, enter BIOS/UEFI setup (usually F2 or Del) and temporarily disable **Secure Boot** or enable USB boot. Record any change
Wait for MemTest86 to load and show the main screen

No.	Observable performance criteria	Max marks	Marks awarded	Remarks
2.6	Clicks Write, confirms the warning, waits for completion, and safely ejects the flash drive	1	0	Clicked write confirmed warning
	Sub-total	6	05	

Task 3: Booting from the flash drive (5 marks)

No.	Observable performance criteria	Max marks	Marks awarded	Remarks
3.1	Switches off the test computer and inserts the prepared flash drive	1	0	switched off
3.2	Presses the correct boot menu key at start-up	1	0	use f10 for HP LAPTOP
3.3	Selects the USB flash drive from the boot menu	1	0	selects usb flash drive
3.4	Enters BIOS/UEFI and adjusts settings (Secure Boot / USB boot) only where needed, and records the change	1	0	Enters the UEFI
1.5	MemTest86 loads and the main screen is displayed	1	0	mem test 86 loaded and the main screen displayed
	Sub-total	5	05	

4: Running the memory test (8 marks)

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	Observable performance criteria	Max marks	Marks awarded	Remarks
4.1	Starts the test using the default settings (Enter / Start Test)	1	1	Start the test
4.2	Records the RAM size detected by MemTest86	1	1	Recorded RAM size
4.3	Records the RAM type and speed (DDR3, DDR4, etc.)	1	0	DDR Recorded
.4	Notes the test number, pass number, and elapsed time	1	0	Not Recorded
5	Lets the test run without interrupting, touching, or powering off the computer	1	1	
	Allows at least 1 full pass to complete	2	2	
	Monitors the Errors counter and records the number of errors and the test during which they occurred	1	1	Monitors the error counter
	Sub-total	**8**	06	

Results and report (6 marks)

)bservable performance criteria	**Max marks**	**Marks awarded**	**Remarks**
ads and records the final result correctly (SS or FAIL)	1	1	Reads final result as Pass
s MemTest86 correctly using Esc / Exit	1	1	Exits memtest

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1 If the computer will not boot from the USB, enter BIOS/UEFI. Disable **Secure Boot** or enable USB boot. Record any changes you make below the main screen.

No.	Observable performance criteria	Max marks	Marks awarded	Remarks
5.3	Removes the flash drive and restarts the computer normally	1	1	Restarts pc
5.4	Locates and views the report file on the USB (EFI\BOOT folder) using a working computer	1	01	Not reached
5.5	Fills in the answer sheet as the work progresses (not from memory afterwards)	1	01	filled
5.6	Observes safety and procedure throughout (no unplugging or powering off during the test, no unauthorised BIOS changes)	1	01	Observed
	Sub-total	**6**	05	

Practical checklist summary

Task	Max marks	Marks awarded
Task 1: Preparation and download	5	05
Task 2: Creating the bootable flash drive	6	05
Task 3: Booting from the flash drive	5	05
Task 4: Running the memory test	8	06
Task 5: Results and report	6	05
TOTAL PRACTICAL	**30**	26

SECTION C: PRODUCT CHECKLIST (5 MARKS)

Assess the finished products submitted by the candidate.

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No.	Product criteria	Max marks	Marks awarded	Remarks
1	Bootable MemTest86 flash drive is functional and loads MemTest86 on the test computer	1	0	Bootable memtest86 functional
2	Answer sheet is fully completed (all 10 entries filled in)	1	0	incomplete
3	RAM details recorded (size, type, speed) are accurate and match the computer tested	1	0	Recorded
4	Test results recorded (passes, time, errors, PASS/FAIL) are consistent with the MemTest86 report	1	0	Results Recorded
5	Conclusion on RAM health is logical and gives a correct recommendation (for example reseat the RAM, test one stick at a time, or replace the module if errors are found)	1	0	No Recommendation
	TOTAL PRODUCT	5	0/3	

Oral questions checklist(30 marks)

No.	Oral question	Expected answer	Max marks	Marks awarded	remarks
1	What does POST stand for?	Power-On Self-Test.	2	02	Stated correctly

If the computer will not boot from the USB, enter BIOS/UEFI setup (usually **F2** or **Del**) and temporarily disable **Secure Boot** or enable USB boot. Record any change you make.
Wait for MemTest86 to load and show the main screen.

2	Why does a computer run POST when it is switched on?	To check that the main hardware is present and working before the operating system loads.	2	0	incomplete
3	Name two parts of the computer that POST checks.	Any two of: CPU, RAM, video card, keyboard, storage (hard drive/SSD), motherboard.	2	02	RAM card drive started
4	What does one short beep usually mean?	POST passed (completed successfully).	2	002	~~incorrect~~ given correctly
5	What does BIOS stand for, and what is UEFI?	Basic Input/Output System. UEFI is the modern replacement for BIOS.	2	0	
6	What is the CMOS battery used for?	It keeps the BIOS settings and the clock (date and time) when the computer is switched off.	2	0	incomplete
7	The computer has no beep, no display, and no fan. What is the likely problem?	A power problem (power supply, power cable, or power button).	2	02	power supply not started
8	Where can you find the meaning of a beep code?	In the motherboard or computer manual, or a beep code chart.	2	0	Ganarally one
)	Name two tools used to test a computer that fails POST.	Any two of: POST card, multimeter, PSU tester, known-good parts, motherboard speaker.	2	00	In comect
)	Give two reasons why we test computer components.	Any two of: to find the faulty part, to save time and money, to confirm the repair worked, to	2	02	

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		avoid the fault returning, for safety.			
11	What is swap testing?	Replacing a suspected faulty part with a part that is known to be good, to see if the fault goes away.	2	0	incomplete
12	What is MemTest86 used for?	Testing the RAM (memory) for errors.	2	2	use stated accurately
13	Why do you run MemTest86 from a USB flash drive and not inside Windows?	So Windows does not interfere with the test. It runs outside the operating system.	2	0	Incorrect
14	What should you do if MemTest86 shows errors?	The RAM is faulty. Test one RAM stick at a time to find the bad one, then replace it.	2	0	incorrect response
15	Name one software tool used to check the hard drive, and one used to check the CPU temperature.	Hard drive: CrystalDiskInfo. CPU temperature: Core Temp.	2	1	only crystal disk stated
	TOTAL		**30**	18	

MARY OF SCORES

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Assessment	Max marks	Marks awarded
Practical checklist	30	27 ~~25~~ 26
Product checklist	5	03
Oral checklist	30	18

DECISION OUTCOME

Tick one

☒ ~~Competent (C)~~	☐ Not Yet Competent (NYC)
~~Below 49~~ Above 50	~~Above 50~~ Below 49

Reason for decision / areas needing improvement:

Improve on oral session

ASSESSOR FEEDBACK

General performance Good

CANDIDATE FEEDBACK

Candidate to comment on the assessment, and tick one:

☒ I am satisfied with the assessment process and the decision.

☐ I am not satisfied and wish to appeal.

Comments:

The test was fair.

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TASK 4: Running the assessment test

1. Of

SIGNATURES

	Name	Signature	Date
Assessor	James		02/10/26
Candidate	Collins Nyatho		

The end.